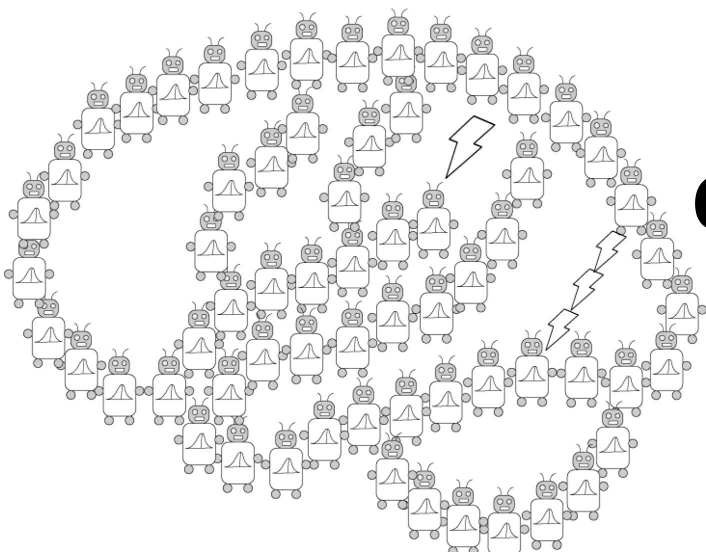
Bayesian Generalization



PSYCH/COMP SCI/BMI 841: Computational Cognitive Science

Joseph Austerweil

February 13, 2025

- Admin Matters
- Bayesian generalization
  - Continuous dimension
  - Discrete hypotheses
  - No free lunch
- **Short rest 1**
- Student presentations:
  - Jess presenting Xu and Tenenbaum (2000)
  - Aakash presenting Navarro et al (2008)
  - Neuki presenting Austerweil & Griffiths (2010)

Administrative Matters

- Back to standard schedule for rest of semester!
- OH back to Tues 2-3pm via zoom
- Presentations submitted by end of presenting day (Thursday at 8:59pm).
  - Sorry for confusion with linked PDF. Should be fixed now.

- Clusters Assignment due Feb 19 8:59pm
- Generalization Assignment due Mar 5 8:59pm
- Project proposal due Mar 12 8:59pm

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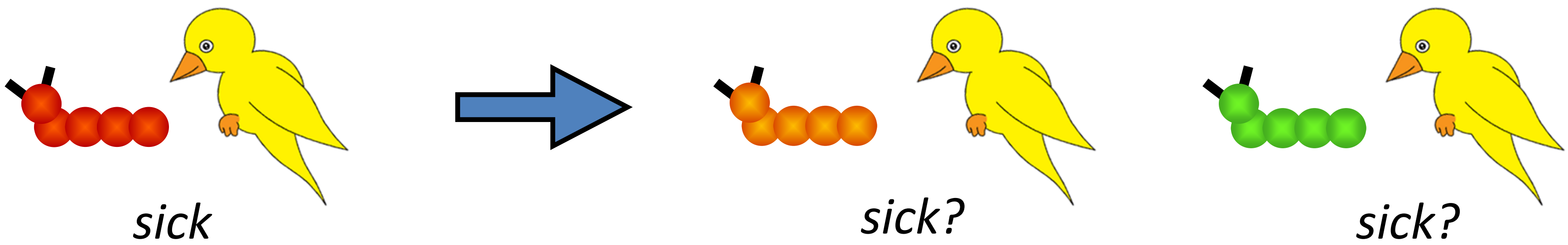
Generalization

No two stimuli are identical. How do we know when to extend a property from previously encountered stimuli to novel stimuli?

This is the problem of generalization (Shepard, 1987).



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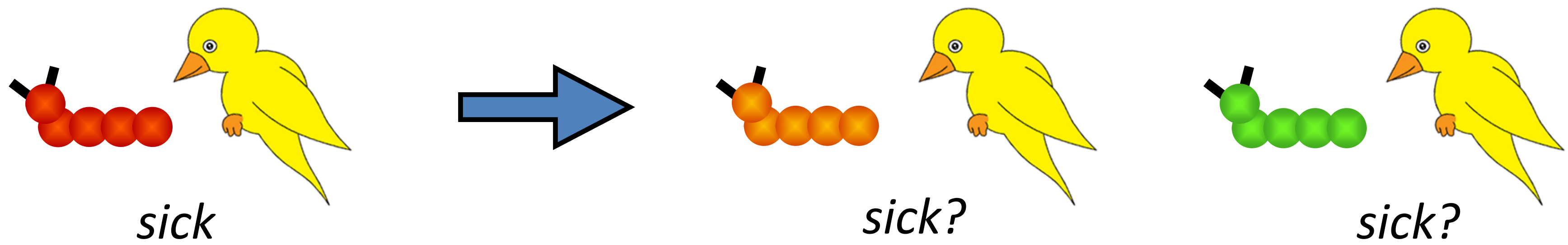
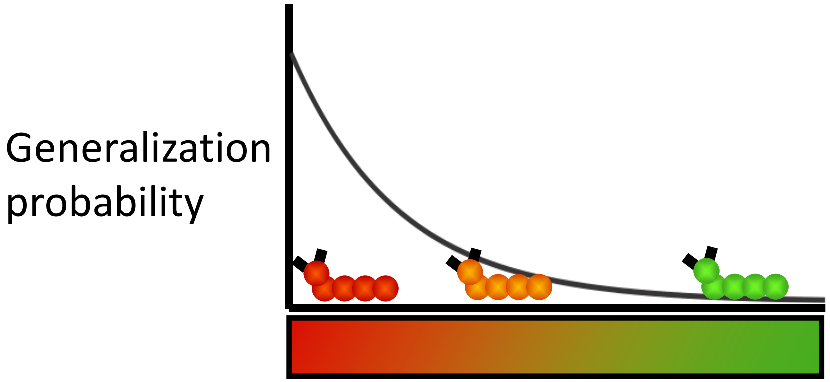


Generalization is pervasive across domains in psychology:
stereotypes, word learning, object recognition, categorization, property induction...



Ideal observer solution: Extend a property from one stimulus to another stimulus to the extent that they are similar or close in psychological space (Shepard, 1987).

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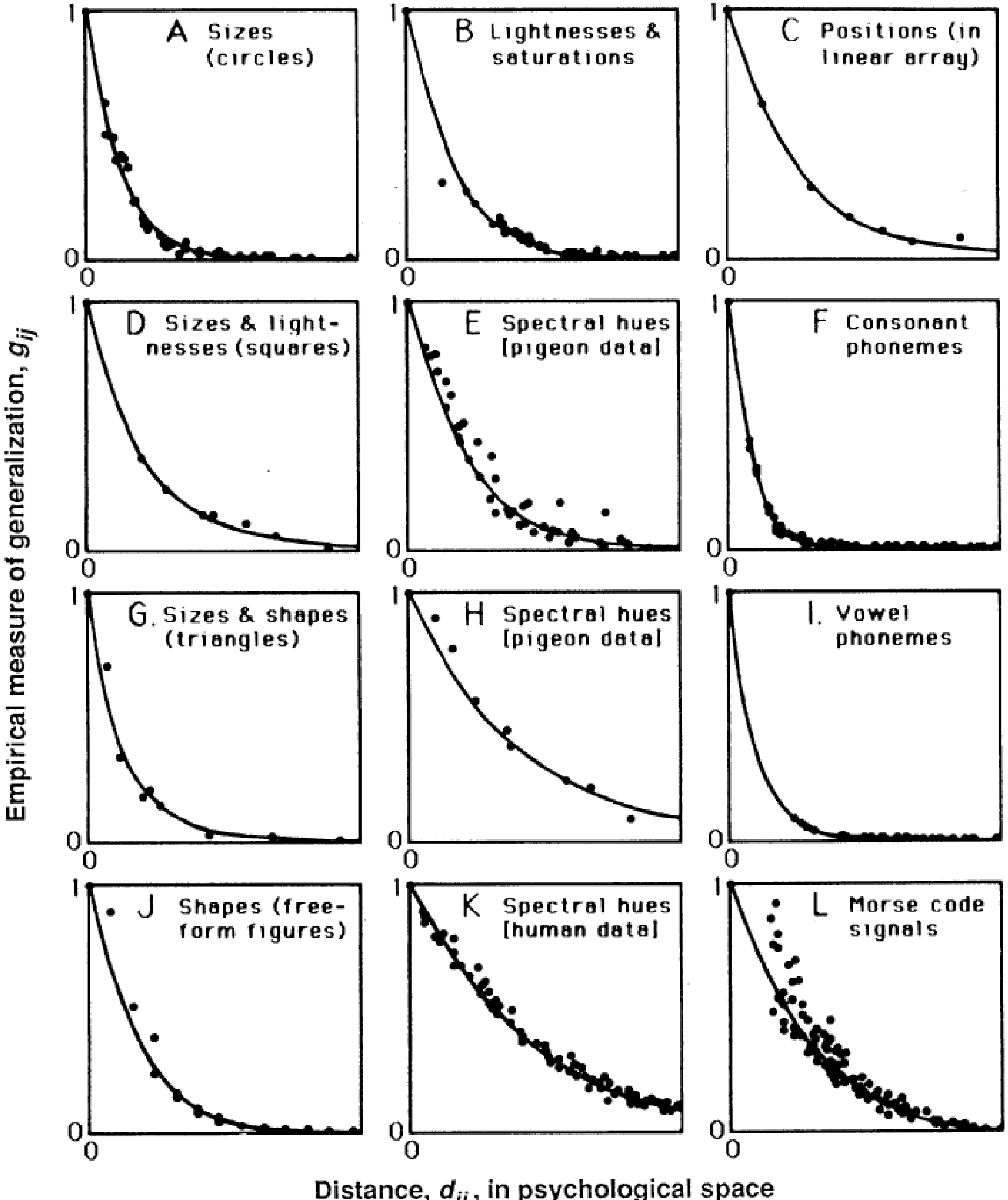
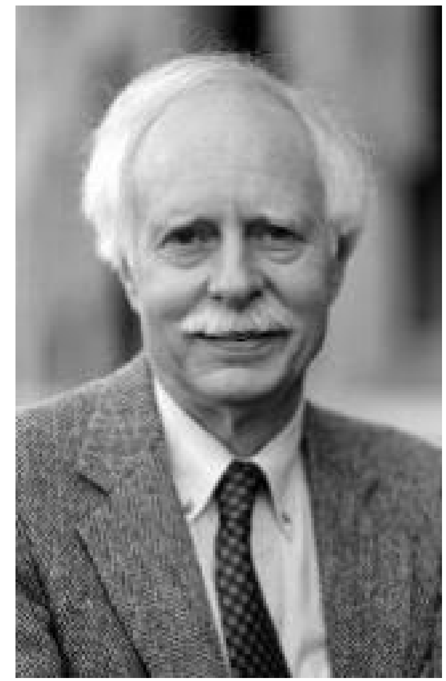


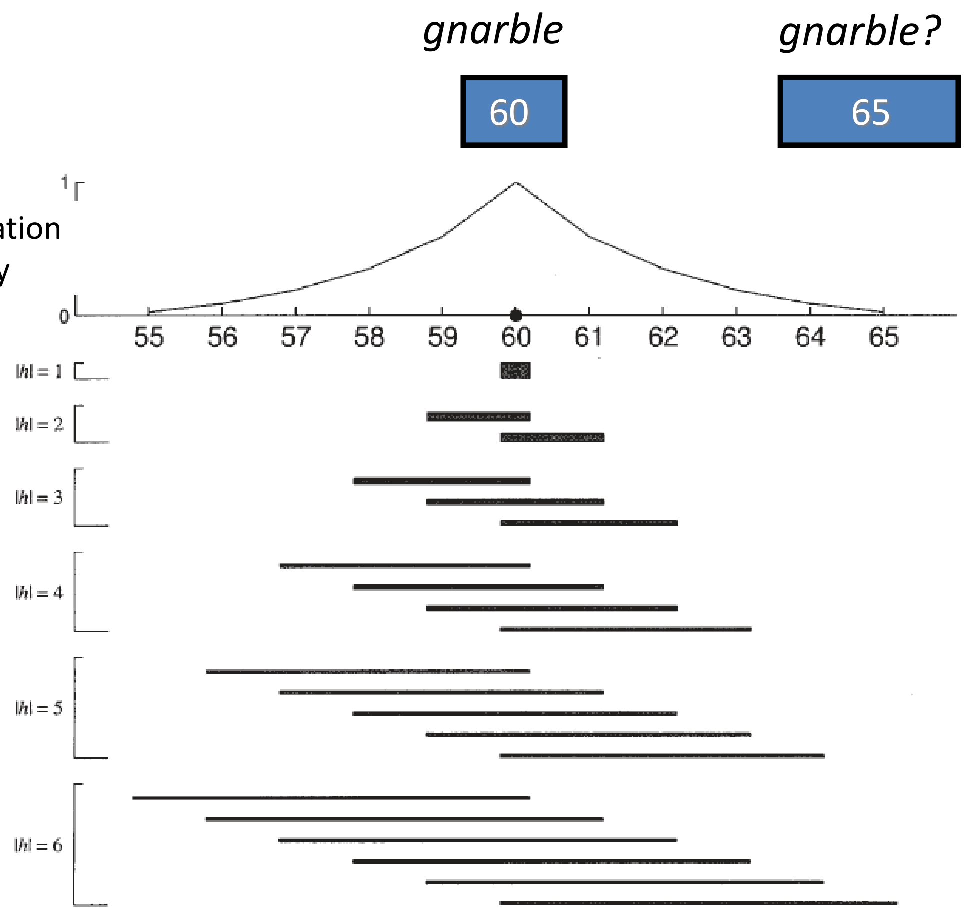


Roger Shepard

Rumelhart prize winner (2006)

(Shepard, 1987)

Computational problem...



Generative process

Gnarbles are rectangles whose widths are in some interval.

$$h = [a, b]$$

Observe a random sample from the set of possible *gnarbles.*

$$P(x|h) = 1/|h|$$

How probable is a given interval to be the set of *gnarbles?*

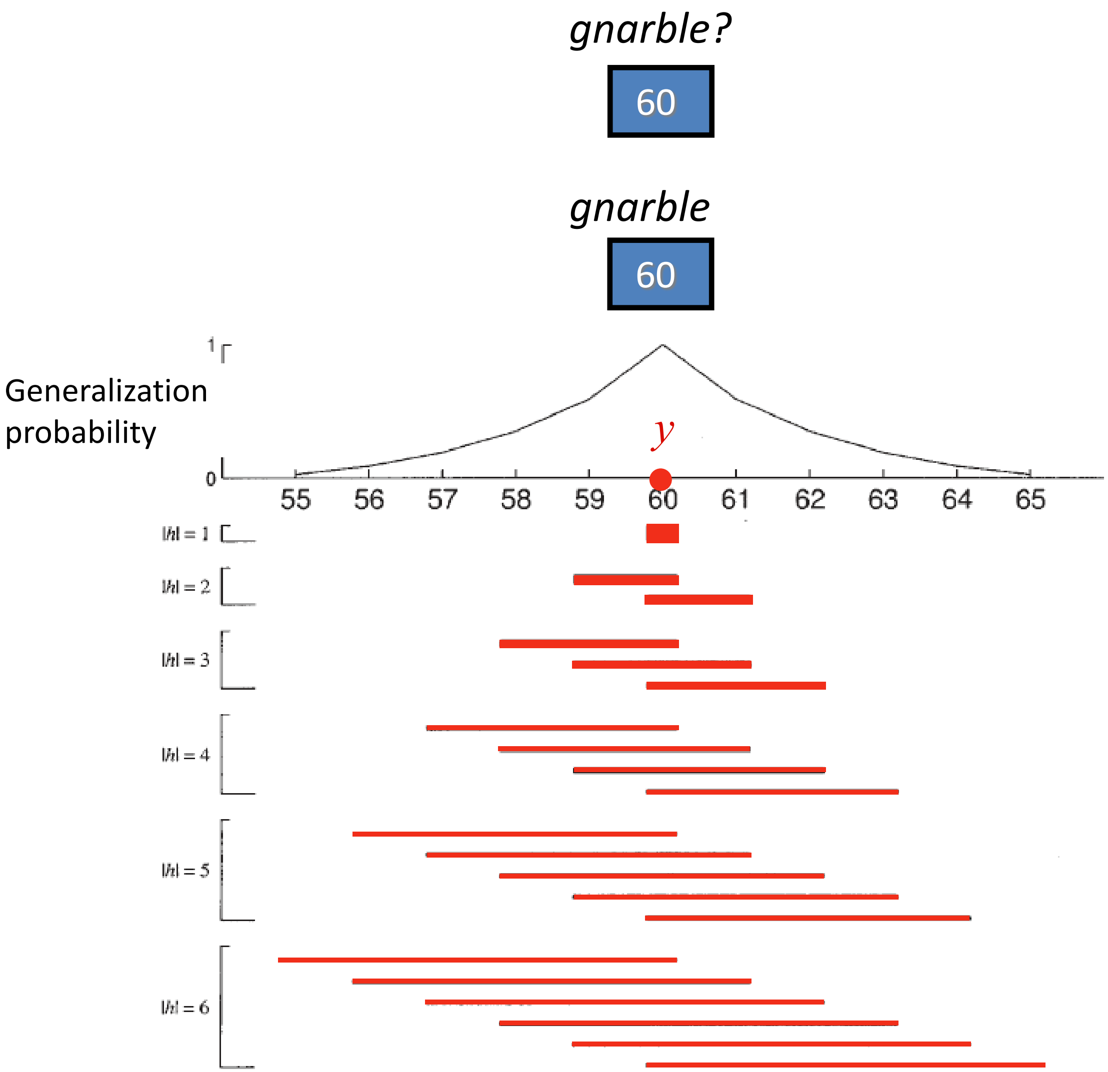
$$P(h|x) \propto P(x|h)P(h)$$

How probable is it that a given rectangle is a *gnarble*?

$$P(y|x) = \sum_{h:y\in h} P(h|x)$$

Shepard (1987), Tenenbaum & Griffiths (2001)

Computational problem...



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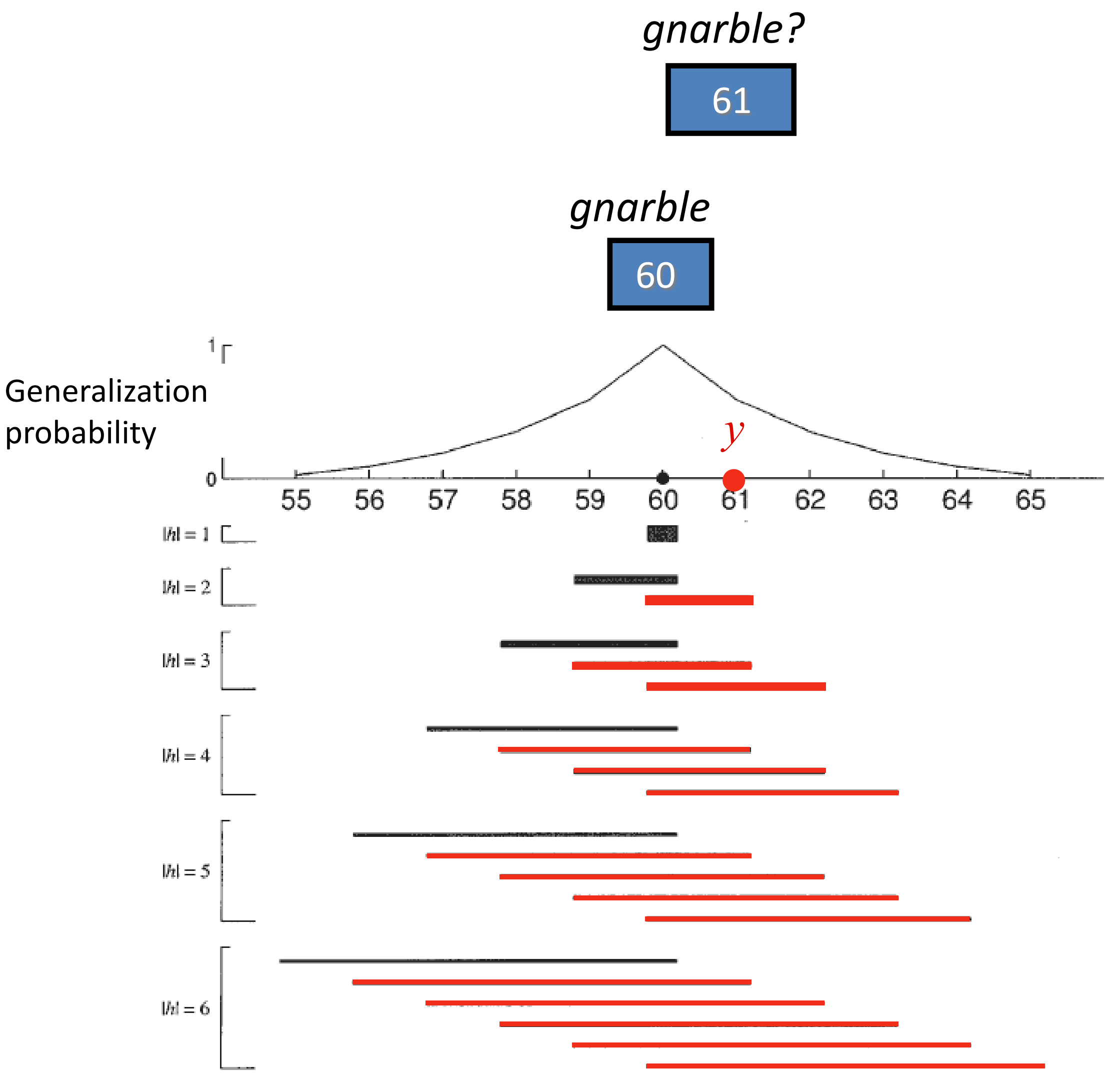

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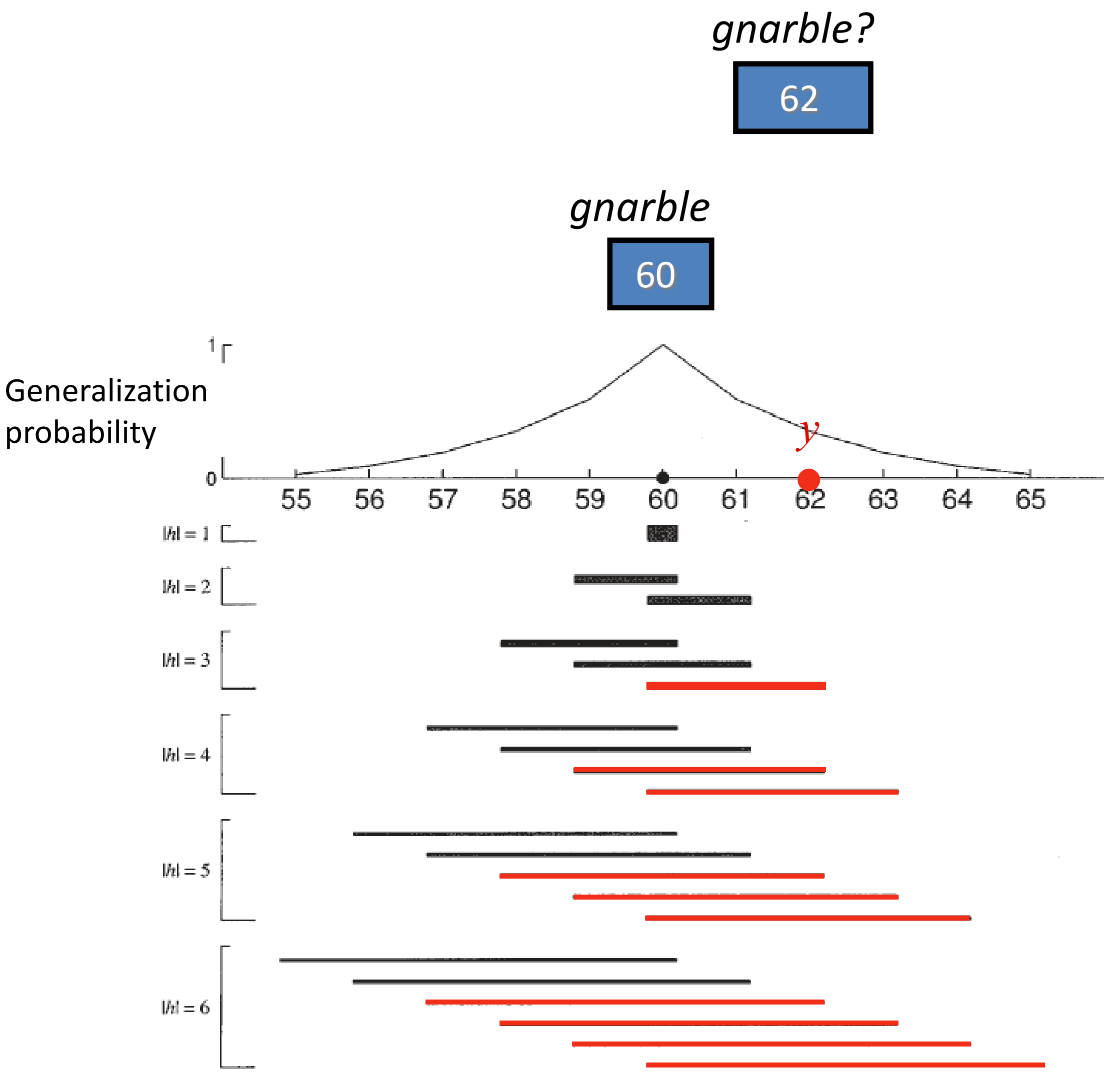

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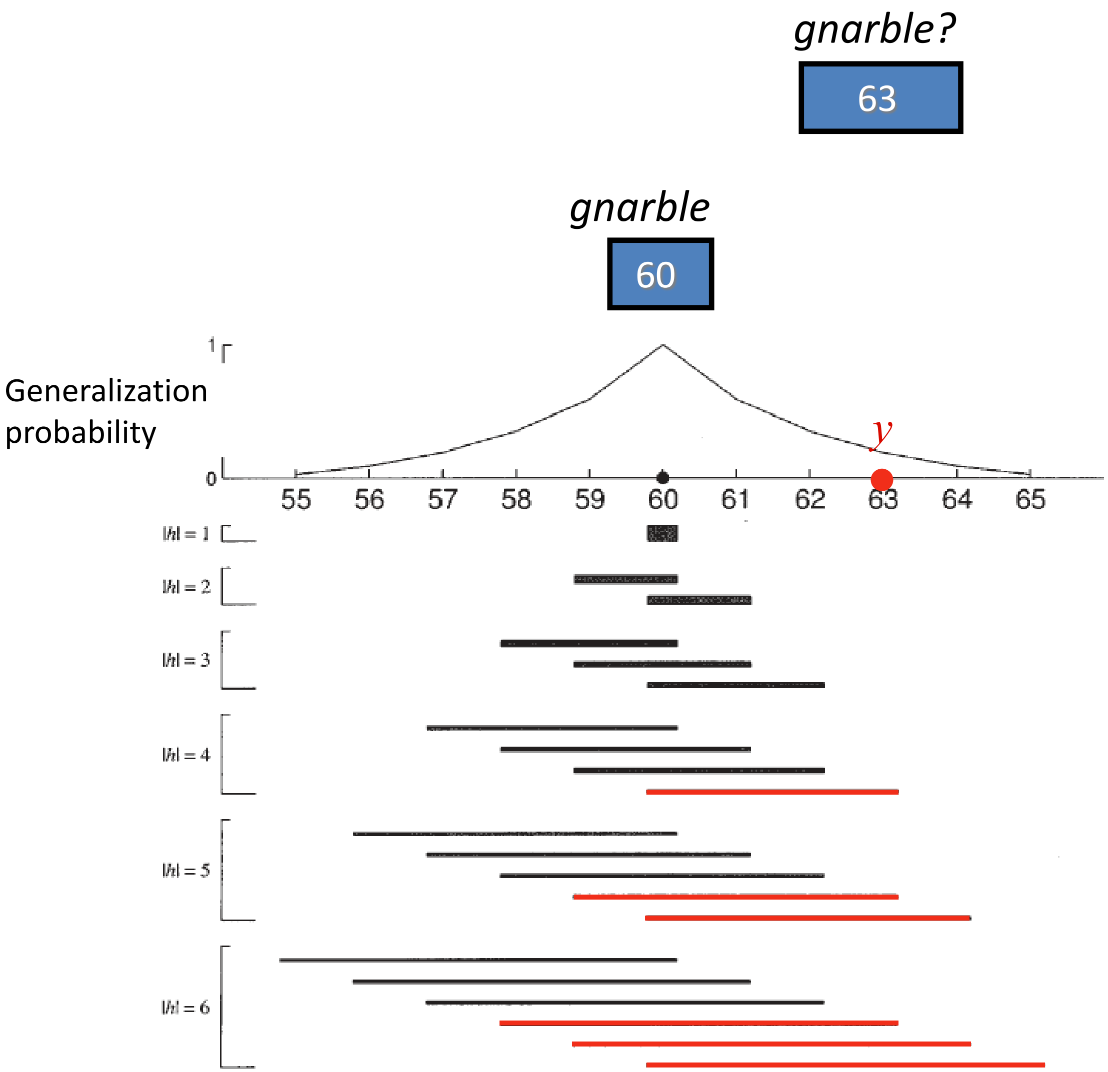

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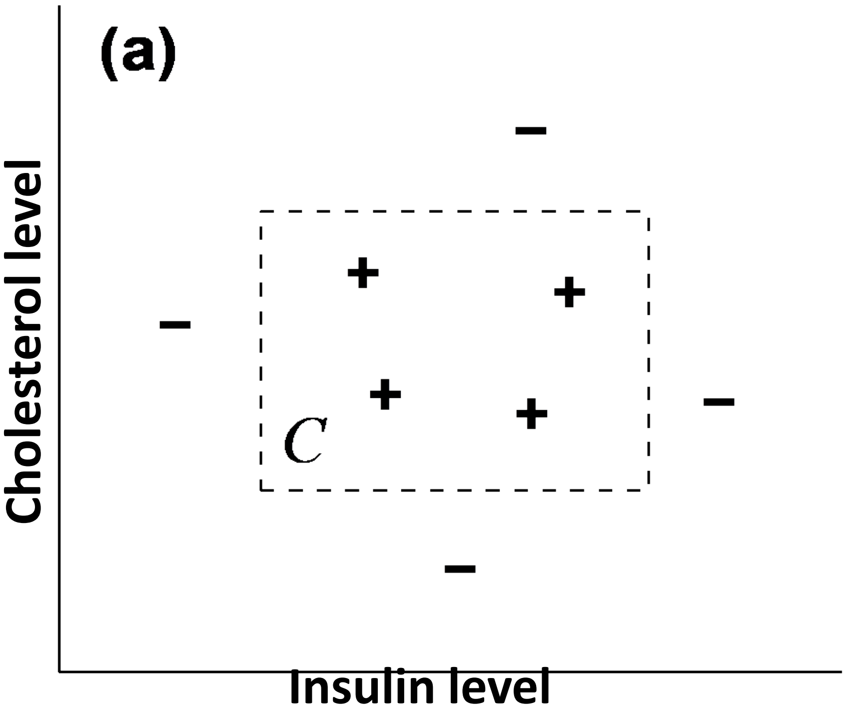
What about 2-D?

2-D Concept



(Tenenbaum, 1999)

Weak v. Strong Sampling

Difference in assumptions that lead to different likelihoods.

Weak sampling: Assume observations randomly sampled and then classified.

$$P(x|h) = \boxed{I(x \in h)}$$

1 if true 0 o.w.

Strong sampling: Assume observation is randomly sampled from the true hypothesis.

$$P(x|h) = \frac{1}{\boxed{|h|}} I(x \in h)$$

Size of hypothesis

(Griffiths & Tenenbaum,2001; Tenenbaum, 1999)

Weak v. Strong Sampling

Weak sampling: Assume observations randomly sampled and then classified.

$$p(y \in C|\mathbf{x}) = \exp\{-(d_1/\sigma_1 + d_2/\sigma_2)\}$$

Distance to closest point

Expected variation on dim 1

Strong sampling: Assume observation is randomly sampled from the true hypothesis.

$$p(y \in C|\mathbf{x}) = [(1 + d_1/r_1)(1 + d_2/r_2)]^{-(n-1)}$$

Distance to closest point

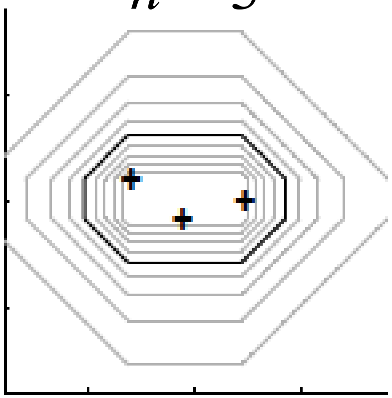
Range on dim 1

(Griffiths & Tenenbaum,2001; Tenenbaum, 1999)

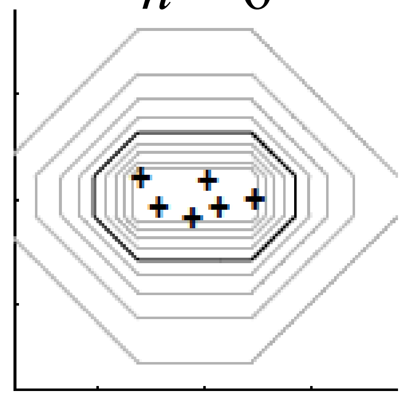
Weak v. Strong Sampling

Weak sampling: Assume observations randomly sampled and then classified.

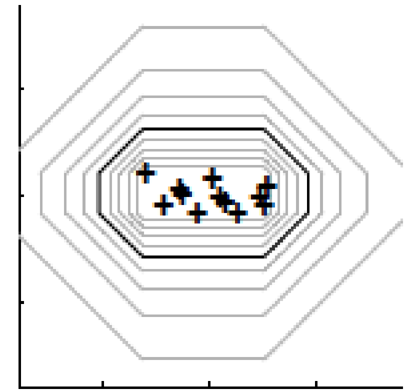
$n = 3$



$n = 6$

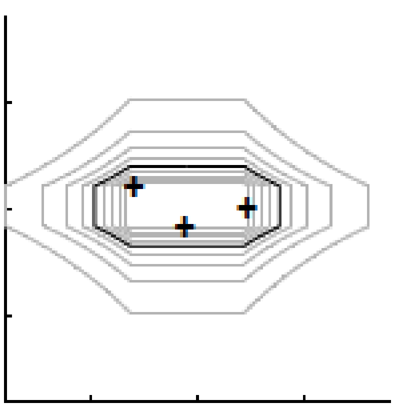


$n = 12$



Strong sampling: Assume observation is randomly sampled from the true hypothesis.

$n = 3$



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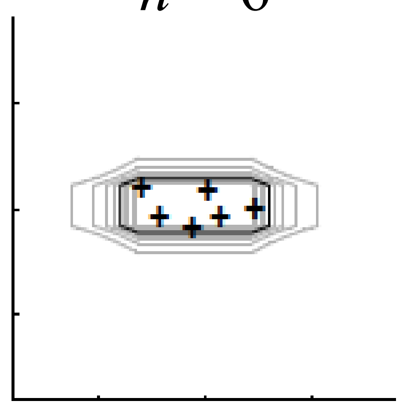

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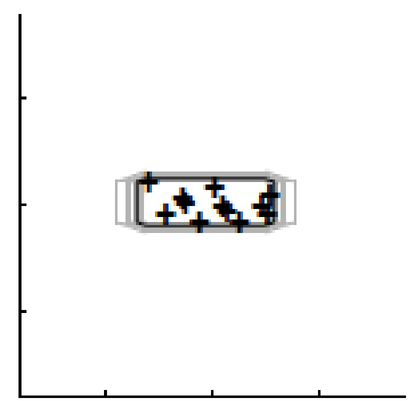


Experiment

Task: Guess 2-D rectangular concepts from examples.

On each trial, several dots from some “arbitrary rectangle of healthy levels” was presented to participants.

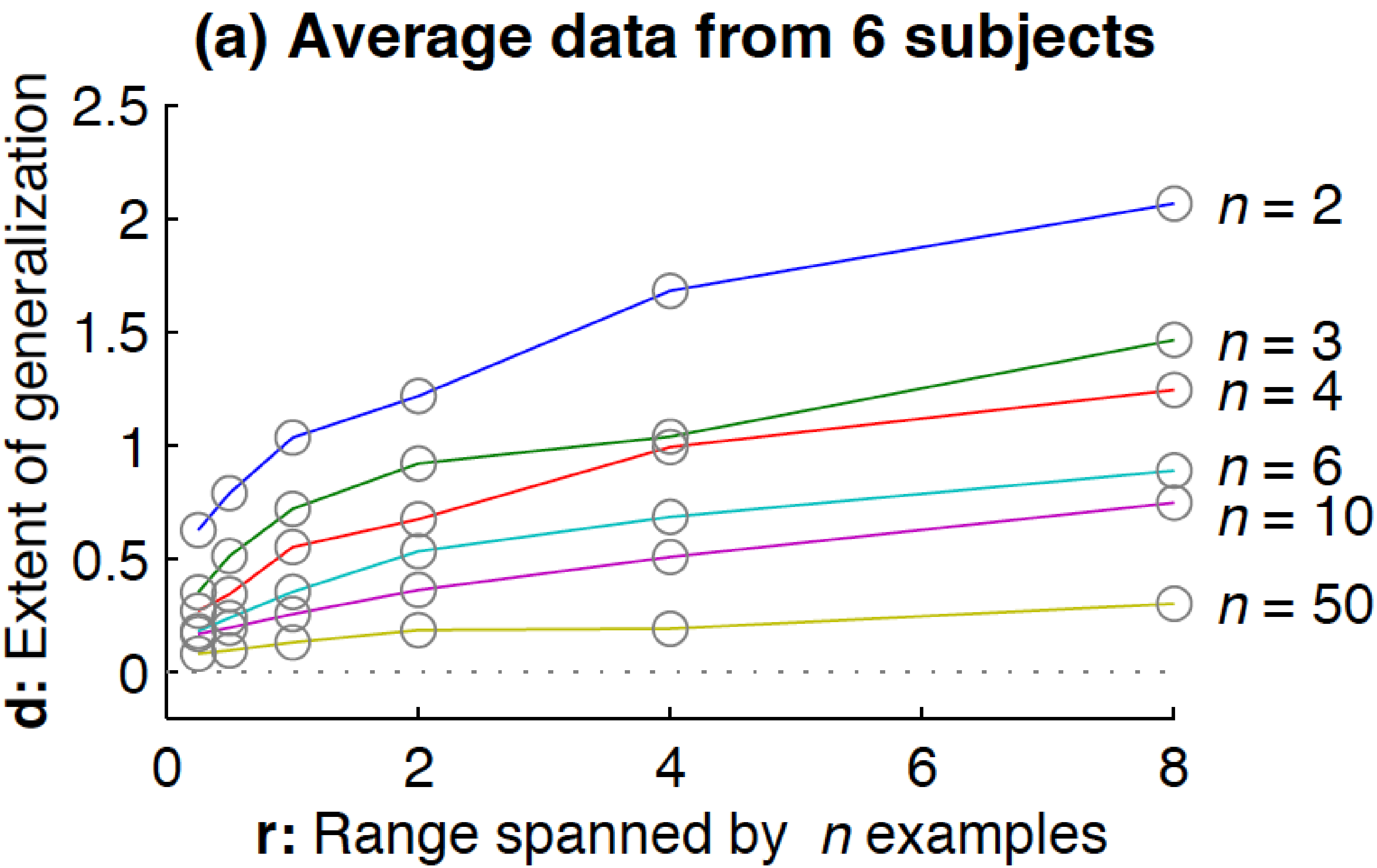
Response: Draw the best guess of the rectangle.

Within-subjects conditions (6 x 6 x 6 = 216 trials):

- Horizontal range: (.25, .5, 1, 2, 4)
- Vertical range: (.25, .5, 1, 2, 4)
- Number of dots: (2, 3, 4, 6, 10, 50)

Empirical Results

Calculated the extent of rectangles d beyond the range of the data r, plotted as a function of r (x-axis) and # of dots n.

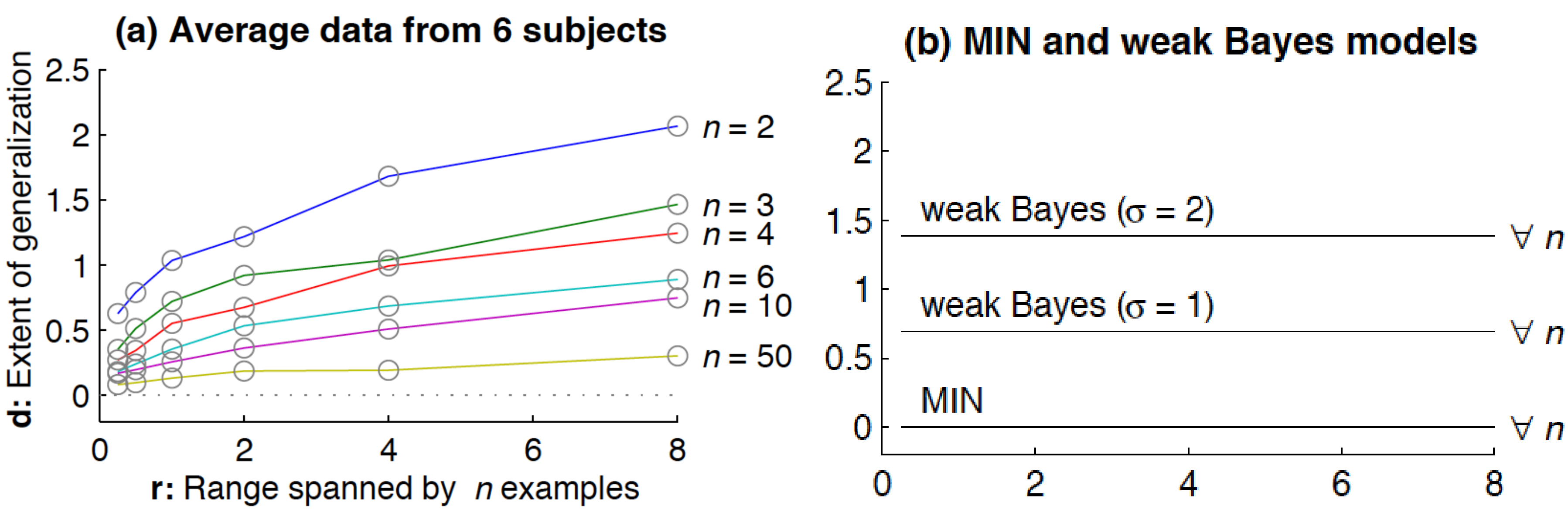


Model Results

Calculated the extent of rectangles d beyond the range of the data r, plotted as a function of r (x-axis) and # of dots n.

Model response: Rectangle of points with probability > 0.5.

Min = tightest rectangle around given points



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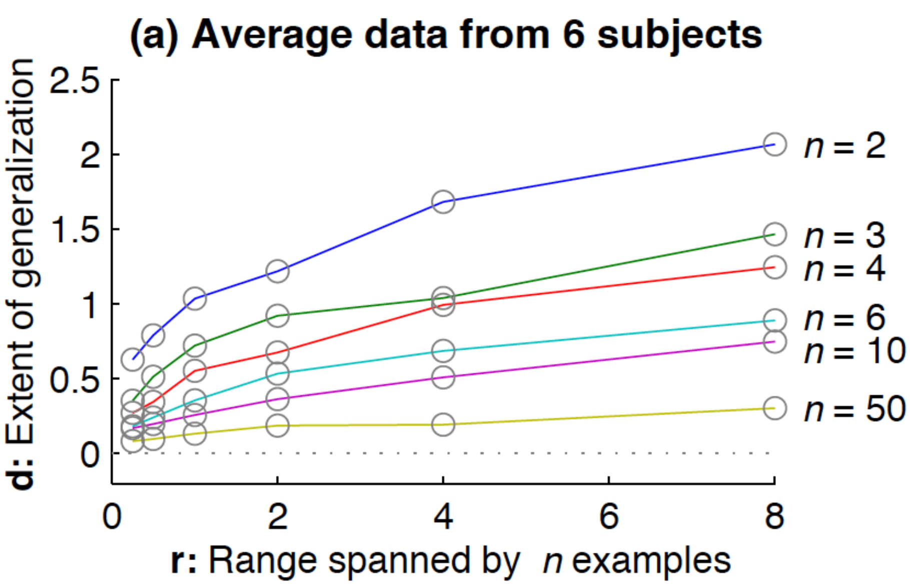
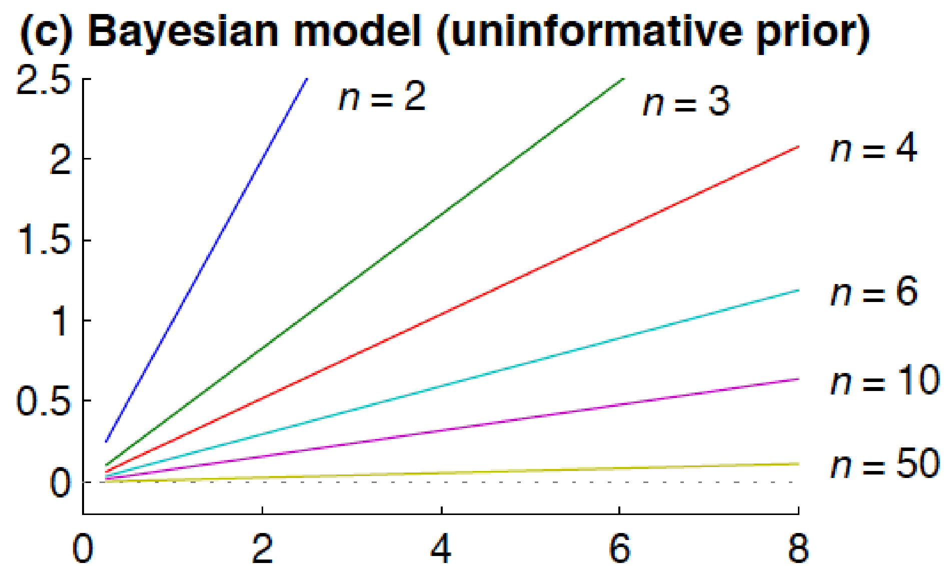




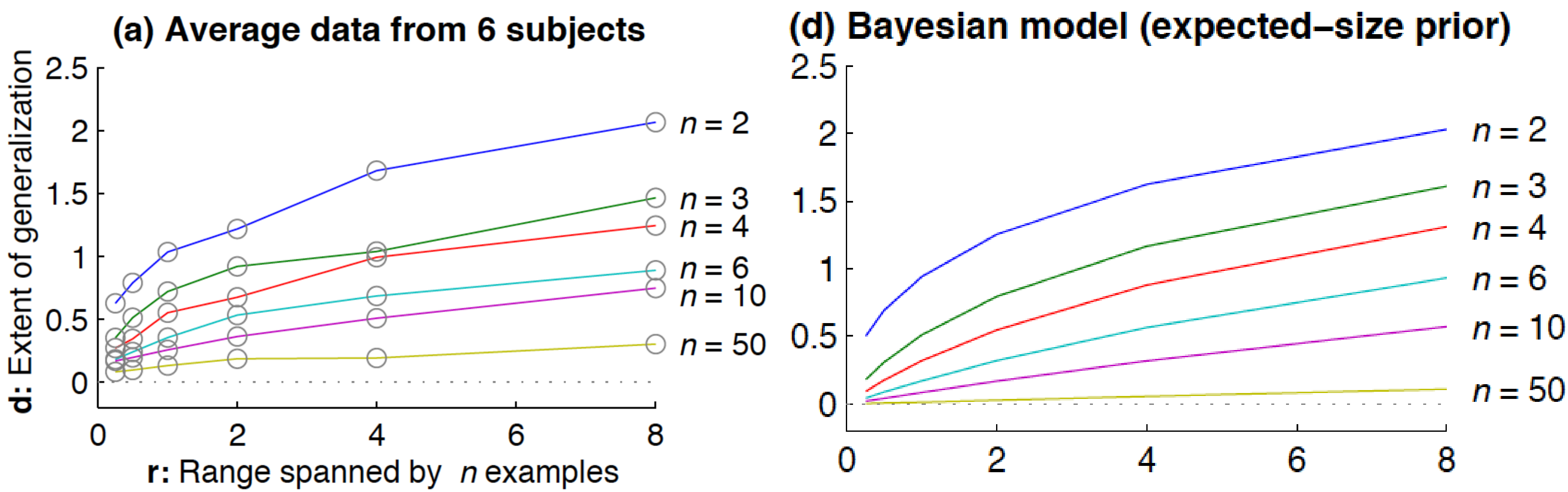
$$p(y \in C|\mathbf{x}) = [1 + d/r]^{-(n-1)} \overset{!}{=} 0.5$$

=> Add a prior!

Model Results

Calculated the extent of rectangles d beyond the range of the data r, plotted as a function of r (x-axis) and # of dots n.

Model response: Rectangle of points with probability > 0.5.



Prior is exponential and justified as due to framing of monitor/prior experience with rectangles.

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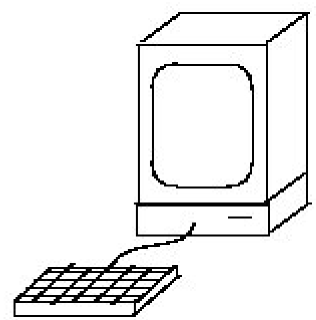
The number game

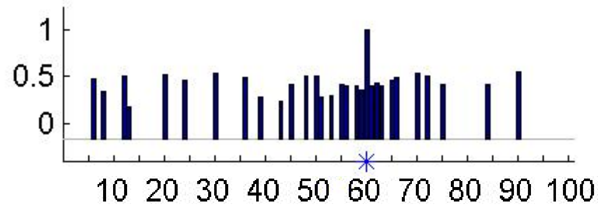


Learning task:

– Observe one or more positive ("yes") examples.

– Judge whether other numbers are "yes" or "no".

The number game

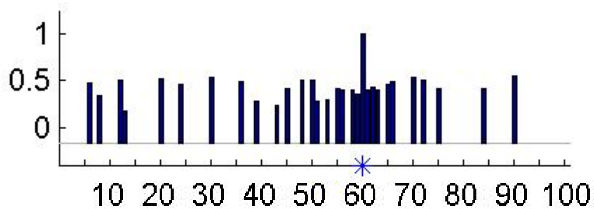
Examples of "yes" numbers	Generalization judgments (N = 20)	
60		Diffuse similarity

The number game

Examples of “yes” numbers

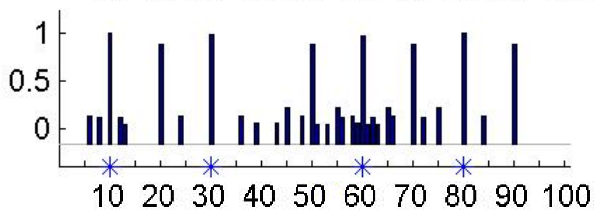
Generalization judgments (N = 20)

60



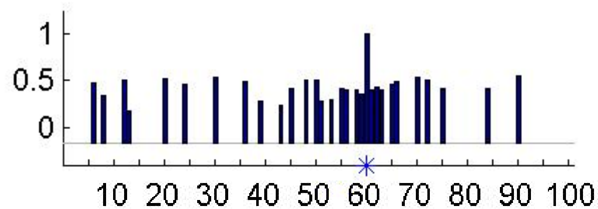
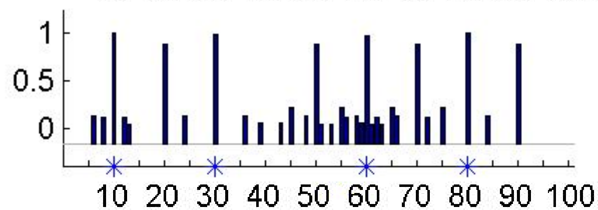
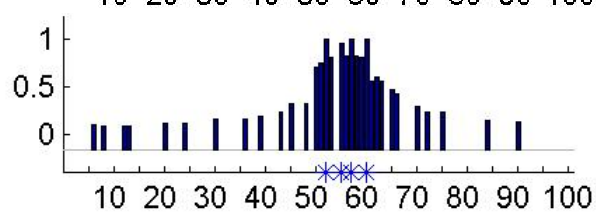
Diffuse similarity

60 80 10 30



Rule:
“multiples of 10”

The number game

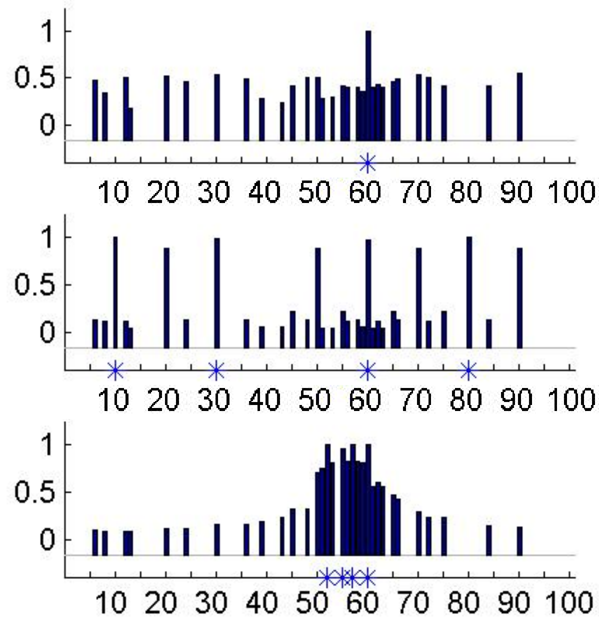
Examples of "yes" numbers	Generalization judgments (N = 20)	
60		Diffuse similarity
60 80 10 30		Rule: "multiples of 10"
60 52 57 55		Focused similarity: numbers near 50-60

The number game

60

60 80 10 30

60 52 57 55



Diffuse similarity

Rule:
"multiples of 10"

Focused similarity:
numbers near 50-60

Main phenomena to explain:

- Generalization can appear either similarity-based (graded) or rule-based (all-or-none).
- Learning from just a few positive examples.

Bayesian generalization framework

(Shepard, 1987; Tenenbaum & Griffiths, 2001)

Generalizing properties as a basis for inductive inference and similarity

h is a property of interest (e.g., "multiple of 10s")

d is observed data (e.g., "8")

$$P(h|x) \propto P(x|h)P(h)$$

Posterior probability

Likelihood

Prior probability

Adapted from Josh Tenenbaum and Tom Griffiths' slides

Bayesian model

H: Hypothesis space of possible concepts:

- h_1 = {2, 4, 6, 8, 10, 12, …, 96, 98, 100} ("even numbers")
- h_2 = {10, 20, 30, 40, …, 90, 100} ("multiples of 10")
- h_3 = {2, 4, 8, 16, 32, 64} ("powers of 2")
- h_4 = {50, 51, 52, …, 59, 60} ("numbers between 50 and 60")
- . . .

Representational interpretations for H:

- Candidate rules
- Features for similarity
- "Consequential subsets" (Shepard, 1987)

Bayesian model

$X = \{x_1, . . . , x_n\}$: n examples of a concept C.

Evaluate hypotheses given data:

$$p(h \mid X) = \frac{p(X \mid h)\, p(h)}{\sum_{h' \in H} p(X \mid h')\, p(h')}$$

$p(h)$ [prior]: domain knowledge, pre-existing biases

$p(X|h)$ [likelihood]: statistical information in examples.

$p(h|X)$ [posterior]: degree of belief that h is the true extension of C.

Prior: $p(h)$

Choice of hypothesis space embodies a strong prior: effectively, $p(h) \sim 0$ for many logically possible but conceptually unnatural hypotheses.

Prevents overfitting by highly specific but unnatural hypotheses, e.g. "multiples of 10 except 50 and 70".

$p(h)$ encodes relative weights of alternative theories:

H: Total hypothesis space

$p(H_1) = 1/5$ | $p(H_2) = 3/5$ | $p(H_3) = 1/5$

H_1: Math properties (24)
- even numbers
- powers of two
- multiples of three

....

$p(h) = p(H_1) / 24$

H_2: Raw magnitude (5050)
- 10-15
- 20-32
- 37-54

....

$p(h) = p(H_2) / 5050$

H_3: Approx. magnitude (10)
- 10-20
- 20-30
- 30-40

....

$p(h) = p(H_3) / 10$

Likelihood: $p(X|h)$

For strong sampling, Size principle: Smaller hypotheses receive greater likelihood, and exponentially more so as *n* increases.

$$p(X \mid h) = \left[\frac{1}{\text{size}(h)}\right]^n \text{ if } x_1, \ldots, x_n \in h$$

$$= 0 \text{ if any } x_i \notin h$$

Follows from assumption of randomly sampled examples and/or law of "conservation of belief":

$$\sum_{\text{all } x \in D} p(D = x \mid h) = 1$$

Captures the intuition of a "representative" sample.

Illustrating the size principle

h_1

h_2

2	4	6	8	10
12	14	16	18	20
22	24	26	28	30
32	34	36	38	40
42	44	46	48	50
52	54	56	58	60
62	64	66	68	70
72	74	76	78	80
82	84	86	88	90
92	94	96	98	100

Multiples of twos:
Each occurs with probability 1/50 = 2%.

Multiples of tens:
Each occurs with probability 1/10 = 10%.

Illustrating the size principle

h_1

Multiples of twos:
Each occurs with probability 1/50 = 2%.

P(60|h) = 1/50

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52	54	56	58	60
62	64	66	68	70
72	74	76	78	80
82	84	86	88	90
92	94	96	98	100

h_2

Multiples of tens:
Each occurs with probability 1/10 = 10%.

P(60|h) = 1/10

Data slightly more of a coincidence under h_1 (5x less likely)

Illustrating the size principle

h_1

h_2

2	4	6	8	10
12	14	16	18	20
22	24	26	28	30
32	34	36	38	40
42	44	46	48	50
52	54	56	58	60
62	64	66	68	70
72	74	76	78	80
82	84	86	88	90
92	94	96	98	100

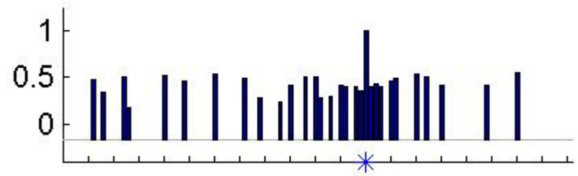
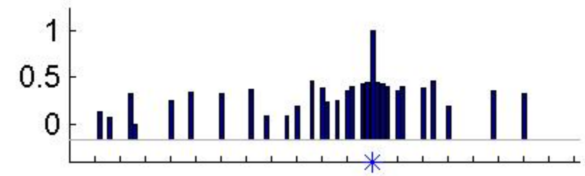
Multiples of twos:
Each occurs with probability 1/50 = 2%.

$P(10,30,60,80|h) = (1/50)^4$
= very small (1.6e-7)

Multiples of tens:
Each occurs with probability 1/10 = 10%.

$P(10,30,60,80|h) = (1/10)^4$
= much less small 1e-4

Data *much* more of a coincidence under h_1 (625x less likely)

+ Examples	Human generalization	Bayesian Model
60		

Building a Bayesian generalization model

1. Pick a domain

Domain of Shapes

- 6 different shapes
- Only differ in "type"

e.g.

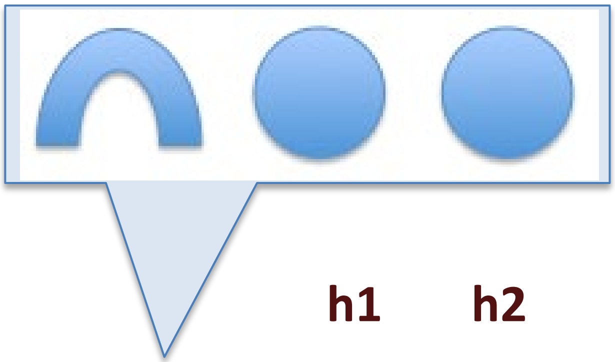
e.g.

✧ presented in <u>triplets</u>, a sequence of three shapes

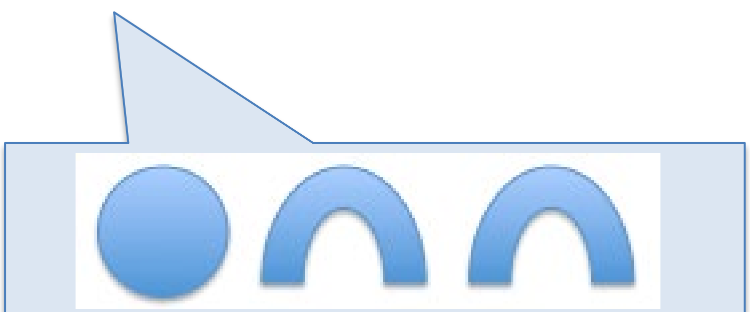
Building a Bayesian generalization model

1. Pick a domain
2. Define hypotheses as a binary matrix, where rows are data points and columns are hypotheses.

Hypothesis Space



	h1	h2	h3	h4	h5	h6
triplet x	1	1	0	1	1	0
triplet y	0	1	0	0	1	1



- h1 = 1st shape is arch
- h2 = follow ABB relation
- h3 = has 1 star
- h4 = has 2 circles
- h5 = has bow(s)
- h6 = 3rd shape is arch

Building a Bayesian generalization model

1. Pick a domain
2. Define hypotheses as a binary matrix, where rows are data points and columns are hypotheses.
   1. If **weak sampling**, done! ($P(x|h)$ = 1 if x is in h)
   2. If **strong sampling**, normalize each hypothesis (divide by the number of examples it has)

   => Elements of the binary matrix encode the likelihood.
3. Pick a prior over hypotheses

   E.g., uniform over hypotheses (each is 1/ # of hypotheses)
4. Do normal Bayesian inference for posterior $P(h|x) \propto P(x|h)P(h)$

   and predictive probabilities $P(y|x) = \sum_{h:y\in h} P(h|x)$

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- Student presentations:
  - Jess presenting Xu and Tenenbaum (2000)
  - Aakash presenting Navarro et al (2008)
  - Neuki presenting Austerweil & Griffiths (2010)

The No Free Lunch Theorem

(Wolpert, 1997)

Averaged over all possible worlds, no learning algorithm is better than any other

– e.g. sequence prediction: given x_1, x_2, predict x_3

Worlds:	Data:	Correct answer:
000	00	0
001	00	1
010	01	0
011	01	1
100	10	0
101	10	1
110	11	0
111	11	1

The No Free Lunch Theorem

(Wolpert, 1997)

In order for a learning algorithm to work better, the

d

You have all the tools now to do Problem Set 2.

You will build your own Bayesian generalization model!

	000		00		0
[redacted]					
	010		01		0
Worlds:	011	Data:	01	Correct answer:	1
	100		10		0
	101		10		1
[redacted]					
	111		11		1

- Admin Matters
- Bayesian generalization
  - Continuous dimension
  - Discrete hypotheses
  - No free lunch
- **Short rest 1**
- Student presentations:
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  - Neuki presenting Austerweil & Griffiths (2010)